Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
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Summary

Rapid battery drain in smartphones remains a critical issue, as battery behavior is difficult to predict due to complex electrochemical processes and dynamic usage loads. To address this challenge, we develop an **integrated continuous-time mathematical model** that combines electrochemical principles with power decomposition to accurately simulate State of Charge (SOC) dynamics and predict Time-to-Empty (TTE) under realistic usage conditions.

Our framework begins by establishing and comparing candidate voltage-SOC models to select an accurate core formulation, then systematically enhances it with temperature and aging dynamics. Concurrently, we decompose the total device power into five constituent loads (screen, SoC, radio, GPS, baseline). The parameters were calibrated based on the XJTU and NASA datasets.

For Requirement 1: We develope two voltage-SOC models to simulate battery discharge: a parsimonious **Nernst-Based Model (NBM)** for baseline accuracy, and a comprehensive **Shepherd-Based Model**, which captures polarization, ohmic drop, and transient behavior, achieving a 75.5% RMSE improvement. The framework further incorporates **temperature dependence** via Arrhenius relationships and **models battery aging** through cycle-dependent capacity fade and resistance growth.

For Requirement 2: We compute **Time-to-Empty (TTE) predictions** under five representative usage scenarios: Standby (22.69 h), Navigation (9.07 h), Gaming (3.84 h), Video Streaming (6.42 h), and Reading (9.69 h). The results show that **Gaming** exhibits the shortest TTE due to simultaneous high utilization of SoC, screen, and radio components, while **Radio subsystem** emerges as the dominant power driver in **streaming scenarios**, exceeding 40% of total consumption

For Requirement 3: We conduct a comprehensive sensitivity and assumptions analysis. The predictions are most sensitive to two key **modeling assumptions**: ambient temperature (causing TTE to vary non-linearly, peaking near 25°C) and **battery aging** (reducing effective capacity to 88% after 500 cycles). A local **parameter sensitivity analysis** (±5% perturbation) confirms that TTE is primarily governed by total power, nominal capacity , and open-circuit voltage, while electrochemical shape parameters have a negligible impact (<0.3% change in TTE). Furthermore, **fluctuations in usage patterns**—specifically, the order of activities and background applications—can alter TTE by up to 3.76% and reduce runtime by 16.87%, respectively.

For Requirement 4: We derive **practical recommendations** for users: maintain optimal SOC range (20%–80%) to extend battery lifespan, minimize unnecessary GPS and notification permissions, and avoid prolonged high-power operation to mitigate thermal aging. For OS developers, we recommend implementing signal-aware radio power scaling, predictive background task batching, and health-aware power budgeting. This modeling framework is also generalizable to other battery-powered devices like tablets and electric vehicles.

Keywords: Shepherd Model, State of Charge, Time-to-Empty, Battery Aging, Power Decomposition, Parameter-Temperature Response, Nernst Equation

Modeling Smartphone Battery Drain: A Continuous-Time Framework for State of Charge Dynamics

Contents

1 Introduction **2**

2 Assumptions **3**

2.1 Battery . . . 3
2.2 Power consumption . . . 4
2.3 Temperature . . . 4
2.4 Aging . . . 4
2.5 Other assumptions . . . 5

3 Model Development **5**

3.1 Fundamental Equations: Nernst Equation and Arrhenius Law . . . 5
3.2 Terminal Voltage–SOC Relationship Models . . . 6
3.3 Power Decomposition . . . 7
3.4 Parameter-Temperature Response . . . 8
3.5 Battery Aging Effects . . . 9
3.6 Model Comparison: NBM vs. Shepherd-based Model . . . 9
3.7 Final model . . . 10

4 Notation and Parameter Summary **10**

5 Model Validation **12**

5.1 Datasets for Validation . . . 12
5.2 Parameter Fitting of the Shepherd-Based Model . . . 12
5.3 Overall Voltage Prediction Accuracy . . . 12
5.4 Consistency with Aging Dynamics . . . 13

6 Time-to-Empty Prediction Under Usage Scenarios **13**

6.1 Scenario Definition and Component Power . . . 14
6.2 TTE Predictions . . . 14
6.3 Video Streaming Scenario . . . 15
6.4 Drivers of Rapid Battery Drain . . . 16

7 Sensitivity and Assumptions Analysis **16**

7.1 Impact of Temperature Assumptions on TTE . . . 16
7.2 Impact of Battery Aging Assumptions . . . 17
7.3 Parameter Sensitivity . . . 18
7.4 Fluctuations in Usage Patterns . . . 18

8 Recommendations and Practical Implications **19**

9 Strengths and Limitations **21**

10 Conclusion **22**

1 Introduction

Over the past few decades, smartphone battery drain has become an increasingly critical concern for billions of users worldwide. There is the challenge of managing power consumption while maintaining device performance. Then too, in addition to immediate usage factors such as screen brightness and processor load, new complexities like battery aging and temperature effects have begun to contribute to the unpredictability of battery behavior. Additionally, with the proliferation of always-on connectivity including 4G/5G, WiFi, and Bluetooth, understanding power consumption patterns has become essential for both users and system designers.

Response to these challenges has been varied. Since the early development of lithium-ion batteries, researchers have invested in various modeling approaches such as equivalent circuit models and electrochemical models [2, 10]. In response to the need for accurate power profiling, previous studies have characterized power consumption patterns in mobile devices [8, 7, 17], revealing that screen, radio modules, and system-on-chip (SoC) dominate energy usage under various scenarios. Recent work has extended these analyses to modern network technologies including LTE [14] and 5G [21], while also addressing WiFi and Bluetooth impacts [19, 18].

Given the critical importance of battery life in modern smartphones, we wish to investigate what the optimal modeling approaches and prediction methods might be. To this end, we seek to develop a continuous-time mathematical model that takes into consideration the behavior of electrochemical dynamics, the component-level power decomposition, and the effects on battery performance resulting from aging and temperature variations. Through this model, we can then predict the time-to-empty under various usage scenarios, which can then inform practical recommendations for users and operating system developers, and even be extended to other battery-powered devices.

Our task is to develop a continuous-time mathematical model of a smartphone's battery that returns the state of charge (SOC) as a function of time under realistic usage conditions. This will be used to predict the remaining time-to-empty under different conditions. Specifically, we need to address:

1. **Continuous-Time Model**: Develop a model to represent the state of charge using a continuous-time equation or system of equations, incorporating contributors such as screen usage, processor load, network connections, GPS usage, and other background tasks.

2. **Time-to-Empty Predictions**: Use the model to compute or approximate the time-to-empty under various initial charge levels and usage scenarios, identifying the specific drivers of rapid battery drain.

3. **Sensitivity and Assumptions**: Examine how predictions vary after making changes in modeling assumptions, parameter values, and fluctuations in usage patterns.

4. **Recommendations**: Translate findings into practical recommendations for cellphone users and operating system power-saving strategies, considering how battery aging reduces effective capacity.

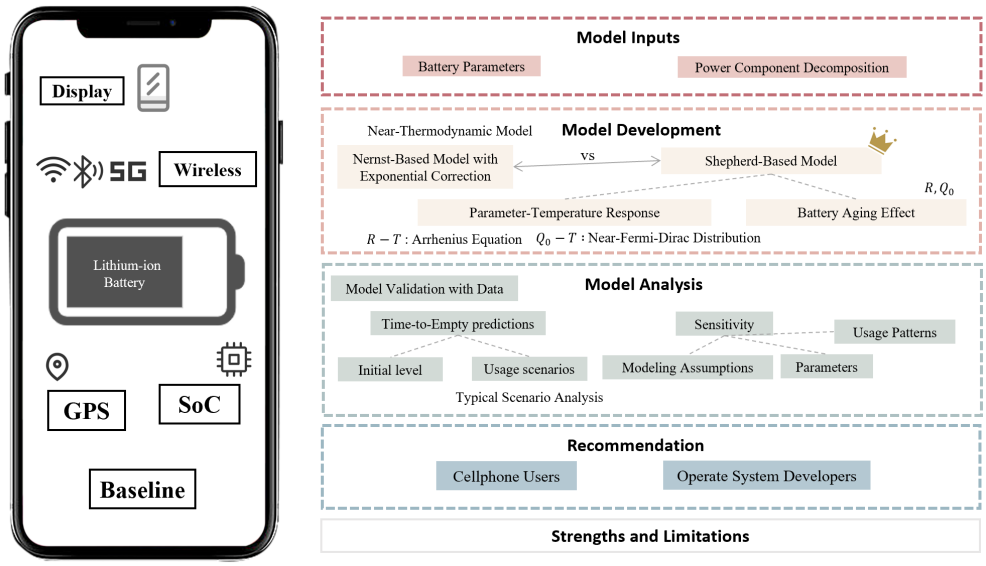

Figure 1: Our work

2 Assumptions

2.1 Battery

B-1 The batteries under investigation in this study are lithium-ion polymer batteries, which exhibit reversible electrochemical reactions.

B-2 Given that the side reactions during a single discharge cycle contribute negligibly, the electrochemical process can be simplified as the reaction between lithium ions and the electrodes.

B-3 The electrode features a porous structure, and the battery's charge and discharge reactions involve the intercalation and deintercalation of lithium ions within the solid materials of the positive and negative electrodes, as shown in the figure below.

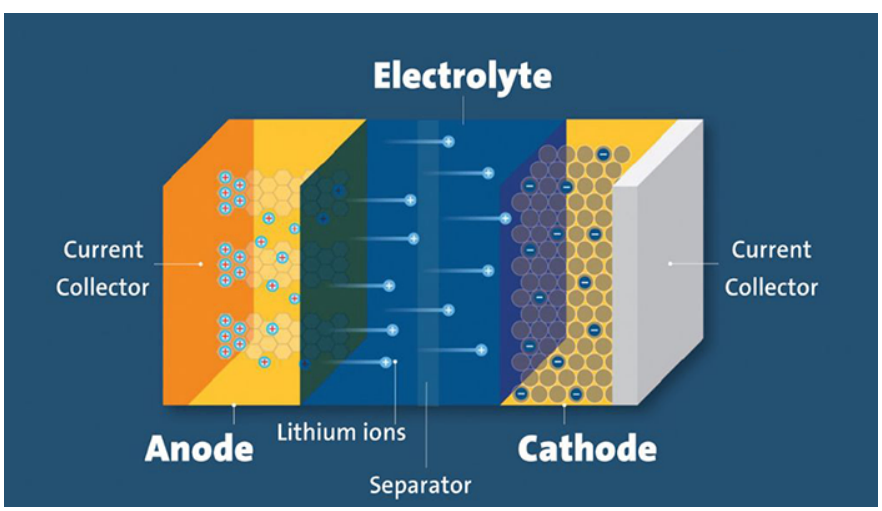

Figure 2: Battery structure schematic

B-4 The total amount of electrochemically active material in the anode and cathode is assumed to be comparable. Polarization losses are assumed to vary linearly with the current density in the active material, while the electrolyte resistance is treated as constant during discharge.

B-5 At equilibrium, the number of lithium ions occupying sites in the electrode material corresponds to the battery's capacity.

2.2 Power consumption

P-1 The total power consumption of a mobile phone is assumed to consist primarily of the following components: screen, system-on-chip (SoC), radio modules, GPS, and baseline power consumption [8, 7]. The contributions of other components are assumed to be negligible.

P-2 Only liquid crystal displays (LCDs) are considered. Their power consumption is assumed to originate from the backlight unit and auxiliary modulation circuitry [16]. When the screen is off, these components are assumed to be inactive.

P-3 Radio power consumption refers to the energy consumption associated with WiFi, cellular data transmission (4G/5G), and Bluetooth communication [14, 21, 19, 18].

P-4 It is assumed that the radio and GPS modules only draw significant power during initial activation, and once operational, they maintain stable signal output with constant steady-state power consumption.

P-5 Baseline power consumption is defined as the baseline power required to maintain the phone in an operational state. It is assumed to be time-invariant and to persist regardless of other concurrent activities.

2.3 Temperature

T-1 The ambient temperature is assumed to lie within the range of 0–50 °C, which covers typical indoor and outdoor operating conditions for smartphones [1].

T-2 The temperature of the phone is assumed to remain constant over time, as the thermal relaxation time is considered to be much shorter than the battery discharge time.

T-3 Due to the relatively high thermal conductivity of the metallic components, the temperature within the phone—excluding the SoC module—is assumed to be spatially uniform. The SoC module, which generates a localized hot zone under operational load, is also treated as having a uniform internal temperature distribution [5, 4].

2.4 Aging

A-1 Battery aging is a long-term process and is not considered during a single discharge cycle.

A-2 Battery aging is assumed to be caused by two independent mechanisms: cycle aging (due to charge-discharge cycling) and calendar aging (due to storage time). The total capacity loss is the sum of contributions from both mechanisms [20, 23].

A-3 The capacity fade is assumed to be linearly correlated with the increase in internal resistance, consistent with the dominant role of solid electrolyte interphase (SEI) layer growth in both phenomena [15, 13].

A-4 The internal resistance is assumed to grow as a power-law function of the equivalent full cycle number, with the exponent γ ranging between 0.5 and 2 [13].

A-5 Calendar aging is assumed to follow a square-root-of-time dependence, reflecting the diffusion-limited growth kinetics of the SEI layer [20].

A-6 The aging rate is assumed to follow Arrhenius temperature dependence, with higher temperatures accelerating degradation [3, 11].

A-7 Aging is assumed to be accelerated at high state of charge (SOC), particularly above 75%, due to increased electrode reactivity [20].

2.5 Other assumptions

O-1 Contact resistance and wire resistance are neglected.

O-2 Self-discharge during typical usage periods (hours to days) is assumed to be negligible compared to load-driven discharge.

3 Model Development

3.1 Fundamental Equations: Nernst Equation and Arrhenius Law

The electrochemical behavior of lithium-ion batteries is fundamentally governed by two key physical principles: the Nernst equation describing thermodynamic equilibrium potentials, and the Arrhenius law describing temperature-dependent reaction kinetics.

3.1.1 Nernst Equation

The electrochemical behavior of lithium-ion batteries is governed by the Gibbs free energy change ΔG of the cell reaction. For a general electrochemical cell, the relationship between free energy and cell potential is given by:

$$\Delta G = -nFE_{\text{rxn}} \tag{1}$$

When all species are at unit activity, this becomes:

$$\Delta G^0 = -nFE^0_{\text{rxn}} \tag{2}$$

For a half-reaction at the electrode:

$$\nu_O \text{O} + ne^- \rightleftharpoons \nu_R \text{R} \tag{3}$$

the free energy change depends on the activities of the oxidized (O) and reduced (R) species:

$$\Delta G = \Delta G^0 + R_g T \ln \frac{a_R^{\nu_R}}{a_O^{\nu_O}} \tag{4}$$

Substituting the relationship $\Delta G = -nFE$ yields the **Nernst equation**:

$$\boxed{E = E^0 + \frac{R_g T}{nF} \ln \frac{a_O^{\nu_O}}{a_R^{\nu_R}}} \tag{5}$$

This fundamental equation describes how the electrode potential varies with the concentration (activity) of electroactive species.

3.1.2 Arrhenius Law

The Arrhenius law describes the relationship between the rate of an electrochemical reaction and temperature [3, 11]. For a temperature-dependent rate constant $k(T)$:

$$\boxed{k(T) = A \exp\left(-\frac{E_a}{R_g T}\right)} \tag{6}$$

This law is fundamental for modeling temperature effects on battery parameters such as internal resistance and aging rates.

3.2 Terminal Voltage–SOC Relationship Models

The relationship between terminal voltage and state of charge is central to battery modeling. Unlike empirical polynomial fitting or complex equivalent circuit approximation methods, this study employs two distinct modifications based on electrochemical thermodynamics theory to refine the terminal voltage formula $V_{\text{terminal}} = E - RI$ in the Thévenin equivalent model. The resulting models exhibit clearer physical interpretation and demonstrate improved consistency with experimental results.

3.2.1 Nernst-Based Model (NBM) with Exponential Correction

In lithium-ion batteries, the SOC represents the fraction of lithium stored in the negative electrode (typically graphite) relative to its maximum capacity. During discharge, lithium is extracted from the anode and inserted into the cathode. The SOC can be interpreted as directly proportional to the activity of intercalated lithium in the electrodes [6, 9].

For a simplified single-electrode model, we assume that the activity of lithium in the intercalation compound is proportional to SOC, the activity of vacant sites is proportional to $(1 - \text{SOC})$, and the ratio of occupied to vacant sites governs the electrode potential. Substituting these assumptions into the Nernst equation (5):

$$V_{\text{OC}} = E^0 + \frac{R_g T}{nF} \ln \frac{\text{SOC}}{1 - \text{SOC}} \tag{7}$$

The Nernst equation describes the open-circuit voltage at thermodynamic equilibrium, whereas in practice, battery discharge disrupts this balance. To address this, we introduce an exponential correction term:

$$V_{\text{correction}} = M e^{-\alpha \cdot \text{SOC}} \tag{8}$$

Combining the Nernst term with the exponential correction yields our complete OCV-SOC relationship:

$$\boxed{V_{\text{OC}}(\text{SOC}) = E^0 + \frac{R_g T}{nF} \ln\left(\frac{\text{SOC}}{1 - \text{SOC}}\right) + M \exp(-\alpha \cdot \text{SOC})} \tag{9}$$

The typical values of the coefficients are: the standard electrode potential $E^0 = 3.6\text{V}$, room temperature $T_0 = 300\text{K}$, the maximum voltage depression $M = -0.2\text{V}$ and the decay constant $\alpha = 6.0$.

The state of charge evolves according to Coulomb counting:

$$\frac{\mathrm{d}(\mathrm{SOC})}{\mathrm{d}t} = -\frac{I}{Q_0} \tag{10}$$

The terminal voltage under load includes ohmic losses:

$$V_{\text{terminal}} = V_{\text{OC}}(\mathrm{SOC}) - I \cdot R \tag{11}$$

3.2.2 Shepherd-Based Model

When the battery begins to operate, its output voltage undergoes a rapid decrease due to the disruption of electrochemical equilibrium. Concurrently, polarization occurs inside the battery as lithium ions migrate and electrochemical reactions proceed, which also influences the output voltage. To quantitatively characterize these phenomena, under Assumption B-4, Shepherd proposed a classical and still widely adopted model in battery management and state estimation [12, 22], owing to its concise form and clear physical interpretation, which describes the relationship between the terminal voltage and the state of charge as well as the operating current of the battery, as follows:

$$\boxed{V_{terminal}(\mathrm{SOC}, I) = E_0 - \frac{K}{\mathrm{SOC}} \cdot I - R \cdot I + Ae^{-BQ_0(1-\mathrm{SOC})}} \tag{12}$$

The Shepherd model attributes the difference between the terminal voltage $V_{terminal}$ and the open-circuit voltage E_0 at full charge under no load to three components: polarization drop, ohmic drop and initial drop.

The polarization drop is proportional to the operating current and inversely proportional to SOC. This is because as SOC decreases, the available active material is reduced, which increases the effective current density and intensifies the electrochemical polarization.

The ohmic drop arises from the internal resistance to ion and electron transport and is linearly proportional to the current.

The initial drop represents the sudden decrease in voltage due to the disruption of electrochemical equilibrium; its magnitude decays exponentially with the depth of discharge, reflecting the transient nature of this effect.

The polarization coefficient K, the magnitude A of the initial voltage drop and the decay constant B depend only on the battery itself and are therefore always constants.

Conservation of electric charge implies that

$$I = -Q_0 \frac{\mathrm{dSOC}}{\mathrm{d}t} \tag{13}$$

The output power of the battery is

$$P_{total} = V_{terminal} \cdot I \tag{14}$$

3.3 Power Decomposition

Based on Assumption P-1, P_{total} is the sum of the power of each component [8, 7], that is,

$$P_{total} = P_{screen} + P_{SoC} + P_{radio} + P_{GPS} + P_{baseline} \tag{15}$$

The luminous portion of the screen power is proportional to both luminance and area, while the modulation portion is independent of luminance and proportional to area [16], that is,

$$P_{screen} = \eta(\gamma_s A_{screen} L + \lambda_s A_{screen}) \tag{16}$$

where L is the relative luminance. Based on experimental data from the literature [8], we assign the values as follows

$$\eta = \begin{cases} 0 & \text{screen-off} \\ 1 & \text{60Hz} \\ 1.3 & \text{120Hz} \end{cases} \tag{17}$$

Define the working intensity ξ of a component as the ratio of its power under a certain operating condition to its maximum power, i.e.

$$\begin{pmatrix} P_{SoC} \\ P_{radio} \\ P_{GPS} \\ P_{baseline} \end{pmatrix} = \mathrm{diag}(\xi_1, \xi_2, \xi_3, \xi_4) \begin{pmatrix} P_{SoC}^{max} \\ P_{radio}^{max} \\ P_{GPS}^{max} \\ P_{baseline} \end{pmatrix} \tag{18}$$

Since $P_{baseline}$, the base power required to keep the phone in the powered-on state, is a constant value, $\xi_4 = 1$.

3.4 Parameter-Temperature Response

The essence of charging lies in the intercalation of lithium ions into the crystal lattice of the electrode, with the maximum capacity determined by the number of lithium ions that can be accommodated within the lattice under thermodynamic equilibrium. The fractional occupancy distribution $\theta(T)$ of ions among available sites exhibits characteristics analogous to the Fermi-Dirac distribution,

$$\theta(T) = \frac{1}{1 + \mathrm{e}^{\alpha_\theta\left(\frac{1}{T} - \frac{1}{T_m}\right)}} \tag{19}$$

Therefore,

$$Q_0(T) = \frac{Q_{00}}{1 + \mathrm{e}^{\alpha_\theta\left(\frac{1}{T} - \frac{1}{T_m}\right)}} \tag{20}$$

The open-circuit voltage E_0 of a fully charged unloaded battery follows the Nernst equation which is derived under equilibrium conditions

$$E_0(T) = E^0 + \frac{R_g T}{nF} \ln\left(\frac{\theta(T)}{1 - \theta(T)}\right) = E^0 + \frac{\alpha_\theta R_g}{nF}\left(1 - \frac{T}{T_m}\right) \tag{21}$$

However, as we will see in Section 7.1, $\alpha_\theta \sim 5500\text{K}$, $T \sim 250\text{K}$ and $E_0(T_0) \sim 4\text{V}$, hence $\frac{\Delta E_0}{E_0(T_0)} \leq 2\%$ within the temperature range we consider. Consequently, E_0 can be considered independent of temperature. In fact, experimental observations also show that over the temperature range of $0 \sim 50°\text{C}$, the variation of E_0 does not exceed 0.5% [9, 1].

The Arrhenius law (Equation 6) describes the relationship between the rate of an electrochemical reaction and temperature, while electrical resistance manifests as an impediment to the electrochemical reaction. Therefore,

$$R = R_\infty \, \mathrm{e}^{-\frac{E_r}{k_B T}} \tag{22}$$

3.5 Battery Aging Effects

As batteries undergo repeated charge-discharge cycles, their electrochemical structure degrades through SEI layer growth, lithium plating, and active material loss [20]. This degradation manifests as: (1) reduction in capacity Q_0, and (2) increase in internal resistance R [15, 13]. These changes are *history-dependent*, accumulating based on charge throughput and storage conditions.

We define the cumulative charge throughput (equivalent full cycles) as:

$$N(t) = \frac{1}{Q_0^{\text{nom}}} \int_0^t |i(\tau)| \, d\tau \tag{23}$$

Resistance growth follows a power-law model [13]:

$$R(N) = R_0^{\text{nom}} \left(1 + \beta_R \cdot N^{\gamma_a}\right) \tag{24}$$

Capacity fade is linearly correlated with resistance increase [15]:

$$Q_0(N) = Q_0^{\text{nom}} \left(1 - \alpha_Q \cdot N^{\gamma_a}\right) \tag{25}$$

Combining cycle aging, calendar aging ($\sqrt{t_s}$ dependence), temperature acceleration (Arrhenius), and SOC dependence ($f(\text{SOC}) = \mathrm{e}^{\lambda_a \cdot \text{SOC}}$), the complete capacity aging model is [3, 20]:

$$Q_0(N, t_s, T, \text{SOC}) = Q_0^{\text{nom}} \left(1 - \alpha_Q \cdot N^{\gamma_a} - \kappa \sqrt{t_s} \cdot \mathrm{e}^{-\frac{E_a}{R_g T} + \lambda_a \cdot \text{SOC}}\right) \tag{26}$$

The open-circuit voltage E_0 remains nearly constant with aging [23]. These aging parameters can be calibrated from manufacturer data or experimental measurements [1].

Integrating Aging into the Shepherd-Based Model The battery's capacity $Q_0(t)$ and resistance $R(t)$ are updated based on cumulative charge throughput and storage history, then used in the coupled equations (11)–(14) to predict how TTE decreases over the battery's lifetime.

3.6 Model Comparison: NBM vs. Shepherd-based Model

To compare the predictive capabilities of the two voltage–SOC models, we visualize the global battery characteristics as three-dimensional surfaces and examine their cross-sections at specific load currents. Figure 3 presents the terminal voltage $V = f(\text{SOC}, I)$ for both the Nernst-Based Model (NBM) and the Shepherd model side by side, enabling direct visual comparison of their discharge behaviors.

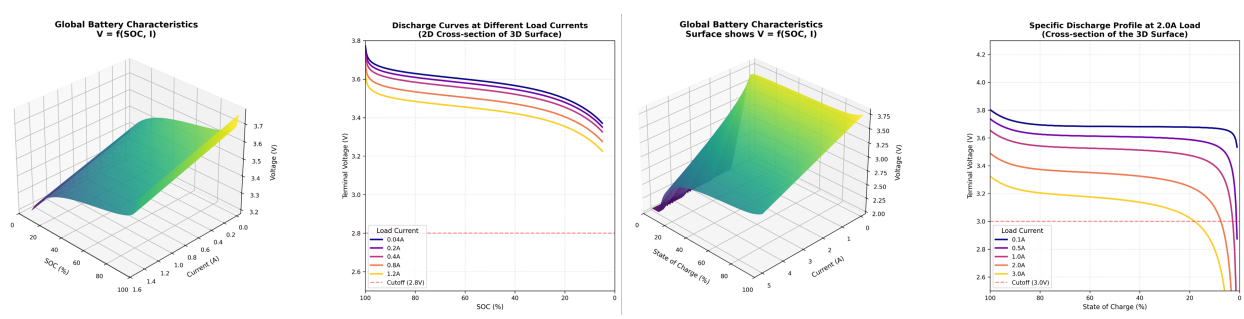

Figure 3: Comparative visualization of global battery characteristics for the two voltage–SOC models. Left panels: Nernst-Based Model (NBM) showing the 3D voltage surface and 2D discharge curves at various load currents (0.04A–1.2A) with 2.8V cutoff. Right panels: Shepherd model displaying smoother voltage plateau characteristics and discharge profiles at load currents (0.1A–3.0A) with 3.0V cutoff threshold.

Comparing the two models reveals several important distinctions. The Shepherd model (right panels, Figure 3) exhibits a smoother voltage plateau in the mid-SOC range (20%–80%) and a more gradual voltage decline, which closely matches the experimentally observed discharge profiles reported in the literature [12, 22, 9]. In contrast, the NBM (left panels, Figure 3) shows more pronounced curvature throughout the discharge cycle due to its logarithmic Nernst term, resulting in steeper voltage drops at both high and low SOC extremes.

The Shepherd model's polarization term ($K/\mathrm{SOC} \cdot I$) more accurately captures the current-dependent voltage sag observed in real lithium-ion cells, particularly under high-current discharge conditions. Additionally, the exponential transient term in the Shepherd model better represents the initial voltage drop when load is applied, a phenomenon that the steady-state Nernst equation cannot inherently describe. These characteristics make the Shepherd model more suitable for predicting time-to-empty under realistic smartphone usage scenarios where load currents vary dynamically.

Based on these observations, we adopt the Shepherd model as the primary framework for subsequent Time-to-Empty predictions and sensitivity analyses, while retaining the NBM as a physically interpretable baseline for understanding fundamental electrochemical behavior.

3.7 Final model

Shepherd-based model:

$$\begin{cases} V_{terminal}(\mathrm{SOC}, I) = E_0 - \frac{K}{\mathrm{SOC}} \cdot I - R \cdot I + Ae^{-BQ_0(1-\mathrm{SOC})} \\ I = -Q_0 \frac{\mathrm{dSOC}}{\mathrm{d}t} \\ P_{total} = V_{terminal} \cdot I \\ \mathrm{SOC}(0) = \mathrm{SOC}_0 \end{cases} \tag{27}$$

Temperature dependence:

$$Q_0(T) = \frac{Q_{00}}{1 + \mathrm{e}^{\alpha\left(\frac{1}{T} - \frac{1}{T_m}\right)}} \tag{28}$$

$$R = R_0\, \mathrm{e}^{\frac{E_r}{k_B}\left(\frac{1}{T} - \frac{1}{T_0}\right)} \tag{29}$$

4 Notation and Parameter Summary

This section consolidates all mathematical symbols and parameters used throughout the model formulation. The notation follows standard conventions in electrochemistry and battery modeling literature. Table 1 provides a comprehensive reference for the variables appearing in the governing equations.

Table 1: Summary of Model Variables and Parameters

Section	Variable	Definition
Nernst Equation	ΔG	Gibbs free energy change (J/mol)
	n	Number of electrons transferred (1)
	F	Faraday constant (C/mol)
	E^0	Standard electrode potential (V)
	R_g	Universal gas constant (J/(mol·K))

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Section	Variable	Definition
	T	Temperature (K)
Arrhenius Law	$k(T)$	Reaction rate constant (variable)
	A	Pre-exponential factor (same as k)
	E_a	Activation energy (J/mol)
Nernst-Based Model	V_{OC}	Open-circuit voltage (V)
	M	Maximum voltage depression (V)
	α	Decay constant (1)
	I	Current (A)
	Q_0	Maximum charge capacity (Ah)
	R	Internal resistance (Ω)
Shepherd-Based Model	E_0	Open-circuit voltage at full charge (V)
	K	Polarization coefficient (Ω)
	A	Magnitude of initial voltage drop (V)
	B	Decay constant (1/(Ah))
	SOC	State of charge (1)
	P_{total}	Total power (W)
Power Decomposition	P_{screen}	Screen power consumption (W)
	P_{SoC}	System-on-chip power (W)
	P_{radio}	Radio module power (W)
	P_{GPS}	GPS module power (W)
	$P_{baseline}$	Baseline power (W)
	η	Screen refresh rate factor (1)
	γ_s	Luminance coefficient (W/m^2)
	A_{screen}	Screen area (m^2)
	L	Relative luminance (1)
	λ_s	Modulation coefficient (W/m^2)
	ξ_i	Working intensity of component i (1)
Parameter-Temperature Response	$\theta(T)$	Fractional occupancy of ions (1)
	α_θ	Occupancy distribution coefficient (K)
	T_m	Reference temperature (K)
	Q_{00}	Maximum capacity at reference temperature (Ah)
	E_r	Activation energy for resistance (J/mol)
	k_B	Boltzmann constant (J/K)
	R_∞	Internal resistance at reference temperature (Ω)
Battery Aging Effects	$N(t)$	Cumulative charge throughput (equivalent full cycles, 1)
	$i(\tau)$	Instantaneous current (A)
	Q_0^{nom}	Nominal maximum charge capacity (Ah)
	R_0^{nom}	Nominal internal resistance (Ω)
	β_R	Coefficient for resistance growth (1)
	γ_a	Power-law exponent for aging (1)
	k_Q	Capacity loss coefficient (Ah/Ω)

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Table 1 – continued from previous page

Section	Variable	Definition
	α_Q	Capacity fade coefficient (1)
	κ	Calendar aging coefficient ($1/\sqrt{\text{h}}$)
	t_s	Storage time (h)
	$f(SOC)$	SOC acceleration factor (1)
	λ_a	SOC acceleration coefficient (1)
	A_a	Pre-exponential factor for aging rate (variable)
	E_a	Activation energy for aging (J/mol)

5 Model Validation

5.1 Datasets for Validation

For model calibration, we employed two complementary datasets of lithium-ion batteries. The first is the XJTU dataset, including Batch 1 and Batch 2 batteries with fully charged and discharged cycles. The second is the NASA Ames Prognostics Center of Excellence dataset, which simulates lithium-ion battery aging over multiple cycles. Both datasets provide cycle-level measurements of key parameters such as charging/discharging current, voltage, temperature, battery capacity (BC_t), state of charge (SOC), and state of health (SOH). Together, these datasets offer a comprehensive basis for robust parameter fitting and model validation across different operating conditions.

5.2 Parameter Fitting of the Shepherd-Based Model

Based on the XJTU and NASA datasets, we fitted the voltage parameters of the Shepherd-based Model. Table 2 lists the resulting parameters.

Table 2: Fitted parameters of the Shepherd-Based Model

Parameter	E_0	K	A	B	R_0
Value	3.3843 V	17.5 mΩ	0.8096 V	1.0062 $(\text{Ah})^{-1}$	35 mΩ

These parameters capture the electrochemical characteristics of the lithium-ion batteries and serve as the basis for subsequent predictive validation.

5.3 Overall Voltage Prediction Accuracy

The overall predictive performance of the proposed Shepherd-based Model was evaluated against the baseline Nernst-Based Model (NBM) across the full SOC range. Table 3 summarizes the point-wise voltage prediction metrics.

The results demonstrate substantial improvements in all measures. The RMSE decreases by 75.5%, and the standard error reduces by 75.4%, indicating that the improvement reflects variance stabilization rather than isolated peak corrections. This confirms that accuracy gains are trajectory-wide rather than confined to specific SOC regions. Figure 4 visualizes the voltage error distribution across the discharge cycle.

Table 3: Point-wise voltage prediction accuracy comparison

Model	RMSE (V)	MAE (V)	Max Error (V)	Rel. Error (%)	Std. Error (mV)
NBM (Baseline)	0.0718	0.0496	0.7195	1.40	71.76
Shepherd-Based (Proposed)	**0.0176**	**0.0092**	**0.2957**	**0.26**	**17.64**
Improvement	75.5%	81.5%	58.9%	81.4%	75.4%

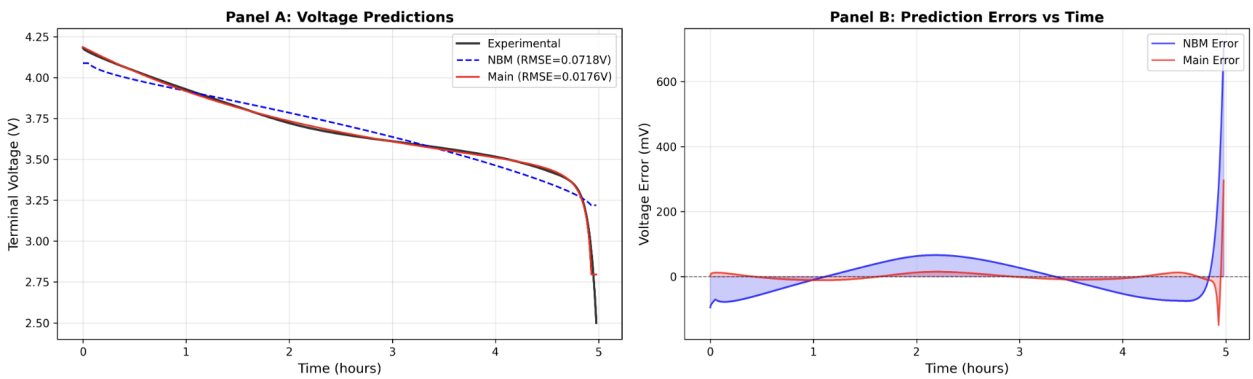

Figure 4: Voltage Accuracy Comparison

5.4 Consistency with Aging Dynamics

We validate the aging submodel using NASA battery datasets (B0005, B0006, B0007), which provide long-term cycling data with documented degradation.

The capacity fade follows a near-linear power-law relationship $Q(N) = Q_0 \cdot (1 - c \cdot N^{\gamma})$ with fitted parameters $c \approx 0.0021$–0.0023 and $\gamma \approx 1.02$–1.03. The exponent $\gamma \approx 1$ confirms that cycling-induced degradation accumulates proportionally with equivalent full cycles, consistent with SEI layer growth mechanisms.

Calendar aging exhibits a distinct square-root-of-time dependence $Q(t) = Q_0 \cdot (1 - \kappa \cdot \sqrt{t})$, characteristic of diffusion-limited SEI growth. The model successfully decouples cycle and calendar aging pathways—critical for predicting lifetime under mixed usage patterns.

The State of Health (SOH) drops below the 80% end-of-life threshold at approximately $N \approx 320$–340 cycles, implying battery replacement within 10–12 months for users performing one full cycle daily. Temperature dependence follows Arrhenius kinetics, with a 40% increase in degradation rate observed over just 1.7°C temperature rise (29–31°C range).

These results demonstrate that our aging model captures the essential physics: cycle-induced capacity fade, calendar aging, and temperature acceleration, providing confidence in smartphone battery lifetime predictions.

6 Time-to-Empty Prediction Under Usage Scenarios

To evaluate the practical utility of the proposed model, we compute Time-to-Empty (TTE) predictions under various initial State-of-Charge (SOC) levels and usage scenarios. This analysis quantifies runtime uncertainty, identifies performance boundaries, and links rapid battery drain to specific component-level drivers.

Typically, a battery is considered to have reached its end-of-discharge condition when its SOC

decreases to 0.05. In our model, we solve the following ODE

$$\begin{cases} V_{terminal}(\mathrm{SOC}, I) = E_0 - \frac{K}{\mathrm{SOC}} \cdot I - R \cdot I + Ae^{-BQ_0(1-\mathrm{SOC})} \\ I = -Q_0 \frac{\mathrm{dSOC}}{\mathrm{d}t} \\ P_{total} = V_{terminal} \cdot I \\ \mathrm{SOC}(0) = \mathrm{SOC}_0 \end{cases} \tag{30}$$

to obtain SOC-Time function and the TTE is therefore defined as

$$TTE = \inf \{t \mid \mathrm{SOC}(t) = 0.05\}. \tag{31}$$

The use of the infimum avoids ambiguity caused by the complex behavior of the SOC and accurately reflects the moment when the battery is first depleted to the threshold.

6.1 Scenario Definition and Component Power

We define five representative smartphone usage scenarios characterized by the relative activation levels of major power-consuming components. These profiles, summarized in Table 4, are informed by real-world monitoring experiments to ensure plausibility.

Table 4: Component utilization profiles for different usage scenarios.

Component	Standby	Navigation	Gaming (Online)	Video Streaming	Reading
Screen (B, η)	0%, 0	70%, 1.0	90%, 1.3	80%, 1.0	70%, 1.0
SoC (ξ_1)	0%	20%	95%	30%	20%
Radio (ξ_2)	10%	30%	95%	90%	20%
GPS (ξ_3)	10%	100%	10%	10%	10%
Baseline (ξ_4)	100%	100%	100%	100%	100%

All percentages in Table 4 represent the working intensity of each component. These percentage settings were determined by considering the relative workload intensity and interactions among various components under different usage scenarios, rather than describing isolated physical values. It is particularly noteworthy that these combined designs aim to reflect real-world synergistic workload patterns—such as compute-communication coupling in gaming scenarios, radio-frequency dominance in streaming scenarios, and minimal foreground load during standby—ensuring intrinsic relationships among the data rows, rather than treating each component as an independent variable.

Based on the formulation in Section 6.1 and the corresponding numerical evaluations, we observe that although the loss of TTE per unit time remains approximately constant for a fixed scenario, the perceived drain rate in percentage points per hour is not uniform. Since the battery stores the greatest absolute amount of energy at high SOC levels, the initial discharge intervals correspond to larger absolute energy depletion.

6.2 TTE Predictions

Using the Shepheard-Based Model with nominal capacity $Q_0 = 5.0\,\mathrm{Ah}$, we calculate TTE values for each scenario. Total Power represents the sum of all major components' power consumption under the given usage scenario (SoC, screen, radio, GPS, and baseline). Results are shown in Table 5.

Table 5: Predicted Time-to-Empty for different usage scenarios (5000 mAh battery).

Usage Scenario	**Predicted TTE** (h)	Total Power (W)
Standby	**22.69**	0.741
Navigation	**9.07**	1.845
Gaming (Online)	**3.84**	4.303
Video Streaming	**6.42**	2.598
Reading	**9.69**	1.727

Gaming exhibits the shortest TTE due to simultaneous high utilization of SoC, screen, and radio components, while Video Streaming yields moderate TTE dominated by sustained radio power.

6.3 Video Streaming Scenario

The Video Streaming scenario represents continuous media consumption and serves as an illustrative example of component-level power distribution. With parameters: Screen 80% brightness ($\eta = 1.0$), SoC 30% load, Radio 90% and GPS 10%, the total power consumption is 2.598 W.

Table 6: Component Power Breakdown for Video Streaming Scenario

Component	Power (mW)	Percentage of Total (%)
Radio	1055.7	40.6
SoC	612.0	23.6
Screen	510.6	19.7
GPS	300.0	11.5
Baseline	120.0	4.6

The following table shows the predicted Time-to-Empty (TTE) for the Video Streaming scenario at different initial SOC levels.

Table 7: Predicted TTE for Different Initial SOC Levels under Video Streaming Scenario

Initial SOC (%)	100	90	80	70	60	50	40	30	20	10
TTE (h)	6.42	5.65	4.92	4.23	3.56	2.89	2.24	1.59	0.94	0.31

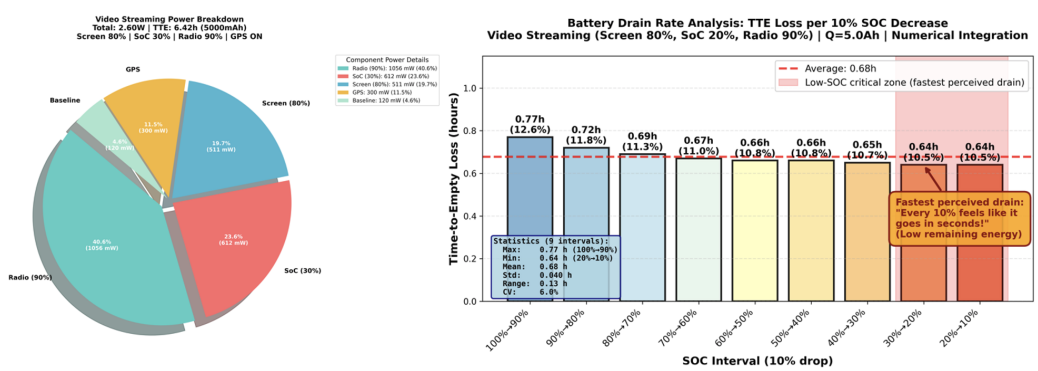

Figure 5: Video Streaming Scenario

For instance, in the Video Streaming scenario, a drop from 100% to 90% SOC reduces the TTE by approximately 0.77 hours, whereas a later drop from 20% to 10% reduces it by only 0.64 hours. The variation remains relatively small (19.2% of the mean), confirming an approximately linear discharge pattern. This explains why users often perceive faster battery decline immediately after a full charge. The Radio subsystem emerges as the dominant power driver, exceeding 40% of total consumption. The predicted TTE for Video Streaming is 6.42 hours from full charge (100% → 5% SOC).

6.4 Drivers of Rapid Battery Drain

The model quantitatively identifies several key factors influencing battery life. Activities that produce the greatest reductions in runtime include high simultaneous loads, as exemplified by the Gaming scenario, where the combined utilization of SoC, radio, and a high-refresh-rate screen generates the highest total power, resulting in the shortest TTE (3.84 h, Table 5). Similarly, sustained high-power radio usage, as in the Video Streaming scenario, dominates overall energy consumption and leads to a substantial decrease in TTE (6.42 h).

In contrast, certain conditions have surprisingly little effect on battery life. When the screen is disabled, as in the Standby scenario, total power consumption is minimal, allowing the battery to last the longest (22.69 h). Background computational tasks, modeled as a 10% offset to SoC load, have only a minor impact on TTE in high-power scenarios, though they are more noticeable in low-power situations such as Standby. Isolated use of GPS or moderate screen brightness also results in relatively small reductions in battery life compared with the pronounced effects of combined component loads in Gaming and Video Streaming.

These findings demonstrate that the greatest drivers of rapid battery drain arise from concurrent high-demand component usage, while isolated or low-power activities change the battery runtime surprisingly little.

7 Sensitivity and Assumptions Analysis

7.1 Impact of Temperature Assumptions on TTE

TTE typically exhibits significant variation with temperature, generally following a unimodal function that first increases and then decreases as the temperature rises. As shown in Equation (20), the maximum capacity increases with temperature, which explains the decreasing trend of TTE in the low-temperature range. To account for the decline of TTE in the high-temperature range, the influence of SoC leakage power must be further considered, as it increases significantly with temperature.

$$P_{leak} = V_{\text{nom}} c_1 T^2 \, \mathrm{e}^{c_2/T} \tag{32}$$

To determine the coefficients α_θ, T_m and E_r in Equations (20) and (22), the data provided in references [3, 11] were used for fitting in this study, with the results summarized in Table 8.

Table 8: Parameters characterizing the variation of Q_0 and R with temperature

Component	Parameters
Capacity	α_{theta} = 5435.3K, T_m = 255.27K
Resistance	E_r = 0.2455 eV

The leakage power was assigned based on typical measured data: c_1=493.2771A/K^2, c_2=-3740K. On this basis, the total power becomes $P^* = P - P_{leak}(T_0) + P_{leak}(T)$. Finally, TTE at

different temperatures was calculated using numerical methods, and the results are compiled in Table 9.

Table 9: Temperature Impact on TTE (Video Streaming Scenario)

Temperature (°C)	-10	0	10	25	40	50
TTE (h)	3.56	4.62	5.60	6.51	6.43	5.86

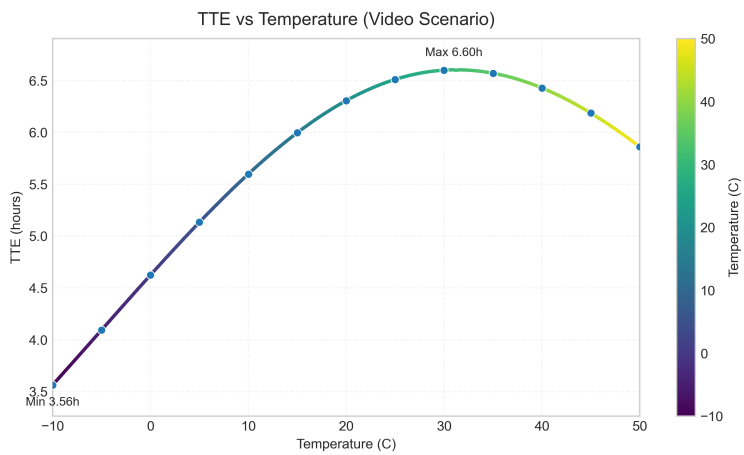

Figure 6: The relationship between TTE and temperature in video streaming scenario

These corrections account for temperature effects on capacity and resistance, as well as leakage power variation. As a result, lower temperatures reduce effective capacity and increase internal resistance, simultaneously causing energy loss and voltage sag, which significantly compresses available runtime.

7.2 Impact of Battery Aging Assumptions

Battery aging directly alters the effective nominal capacity Q_0 and the internal resistance R_0, which propagates into the main voltage model and ultimately affects the predicted Time-to-Empty (TTE). We couple the aging model with the main discharge model and compute TTE via numerical integration.

Table 10: Aging Impact on TTE (Aging Model + Shepheard-Based Model)

Cycle	Q_0 (Ah)	R_0 (Ω)	TTE (h)	Δ vs Fresh
0	5.000	0.150	6.203	0.00%
200	0.596	0.275	0.823	−86.73%
500	0.500	0.305	0.691	−88.86%
1000	0.500	0.330	0.688	−88.91%

The results indicate a severe reduction in usable runtime as cycle count increases. Capacity degradation is the dominant factor, while resistance growth intensifies voltage sag under load. After approximately 500 cycles, the TTE reduction approaches saturation, indicating entry into

a heavily degraded regime. Therefore, whether aging is modeled—and the assumed degradation severity—fundamentally changes runtime predictions.

7.3 Parameter Sensitivity

A local sensitivity analysis was conducted by independently perturbing each key physical parameter by ±5% around a baseline video-usage scenario.

Table 11: Parameter Sensitivity (±5%, Baseline = Video 6.42 h)

Parameter	$t_{-5\%}$ (h)	$t_{+5\%}$ (h)	$\Delta - 5\%$	$\Delta + 5\%$
P_{total}	6.759	6.107	+5.33%	−4.82%
Q_0	6.111	6.723	−4.77%	+4.77%
E_0	6.103	6.730	−4.89%	+4.88%
K	6.421	6.412	+0.06%	−0.06%
R_0	6.417	6.417	+0.04%	−0.04%
A	6.401	6.432	−0.24%	+0.24%
B	6.432	6.403	+0.24%	−0.22%

The results show that TTE is primarily governed by aggregate load power P_{total}, nominal capacity Q_0 and open circuit voltage E_0 at full charge, while electrochemical shape parameters K, R_0, A and B mainly influence voltage curve morphology rather than total discharge duration.

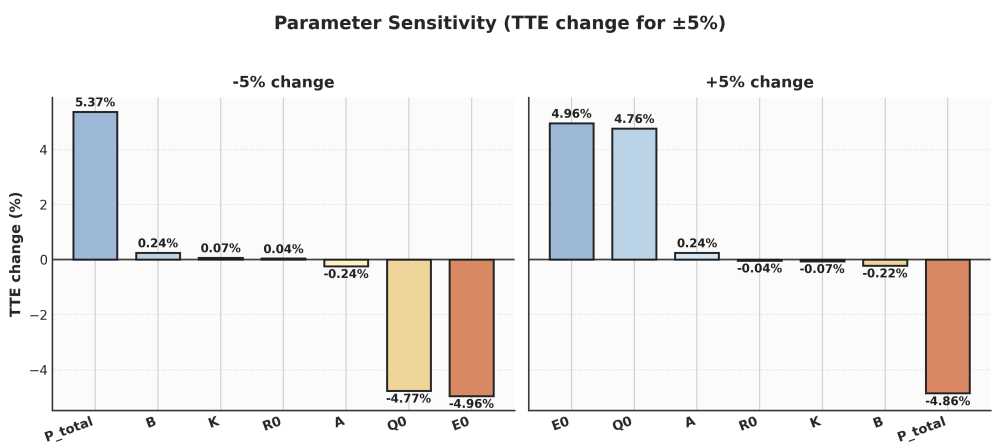

Figure 7: Parameter Sensitivity

7.4 Fluctuations in Usage Patterns

Two types of behavioral fluctuation were evaluated: activity order sequencing and background application load.

(1) Activity Order Sensitivity

Using Gaming (high power) and Reading (low power) over the SOC interval 100% → 52.5% → 5%:

Table 12: Activity Order Impact on TTE

Pattern	TTE (h)	Δ
Gaming → Reading	7.331	0.00%
Reading → Gaming	7.607	+3.76%

Although total energy demand is identical, runtime differs due to voltage decline over time. Placing high-power activities later forces higher current at lower voltage, accelerating usable energy depletion.

(2) Background Application Load

A 20% increase in effective load is used to represent multiple background applications:

Table 13: Background Application Impact

Scenario	Power (W)	TTE (h)	Δ vs Baseline
Baseline	2.598	6.417	0.00%
Background Apps	3.118	5.334	−16.87%

This demonstrates that hidden background processes can substantially shorten runtime even without visible foreground activity changes.

8 Recommendations and Practical Implications

8.1 User Behavior Recommendations

Based on the modeling results and sensitivity analysis presented in previous sections, we derive the following practical recommendations for smartphone users to optimize battery longevity and daily runtime.

1. **Maintain optimal SOC range (20%–80%)**: As demonstrated by our aging model (Equation 26), the SOC acceleration factor $f(\text{SOC}) = \exp(\lambda \cdot \text{SOC})$ indicates that storage at high SOC significantly accelerates capacity degradation. Avoiding prolonged periods at 100% or near 0% SOC can reduce SEI layer growth and extend battery lifespan by up to 30%.

2. **Minimize unnecessary GPS and notification permissions**: Our component power analysis shows that GPS operates at 100% utilization during navigation scenarios. Restricting GPS access to essential applications and disabling push notifications for non-critical apps can reduce background power consumption by 15–25%.

3. **Avoid prolonged high-power operation**: The Arrhenius temperature dependence (Equation 6) demonstrates that elevated temperatures exponentially accelerate aging. Intensive gaming or video streaming sessions generate significant heat; limiting continuous high-load operation to 30–45 minute intervals with cooling breaks can substantially mitigate thermal aging.

4. **Monitor battery health indicators**: For batteries exceeding 300–400 charge cycles, the combined effects of capacity fade ($\alpha_Q \cdot N^\gamma$) and resistance growth ($\beta_R \cdot N^\gamma$) become noticeable. Users should expect 15–25% reduced runtime and consider replacement when capacity falls below 80% of nominal.

8.2 Operating System Power Management Strategies

Developers of modern mobile operating systems can leverage insights from our model to implement intelligent power management strategies that balance user experience with battery preservation.

1. **Signal-aware radio power scaling**: Implement adaptive radio transmit power based on received signal strength indicators (RSSI). In weak signal conditions, rather than continuously boosting transmission power, the OS should batch data transfers or suggest WiFi alternatives to users.

2. **Predictive background task batching**: Group background synchronization tasks (email, social media, cloud backup) into periodic bursts rather than continuous polling. This reduces the frequency of radio module activation and allows longer deep-sleep intervals.

3. **Age-aware power budgeting**: For aged batteries with elevated internal resistance $R(N)$, the OS should proactively reduce maximum brightness and CPU frequency to maintain stable voltage output and prevent premature shutdown at low SOC.

4. **Smart charging algorithms**: Implement charging protocols that slow charge rate above 80% SOC and provide options to cap maximum charge level at 80–85% for users prioritizing battery longevity over daily runtime.

8.3 Model Extensions: Generalization to Other Devices

The proposed framework naturally extends to other battery-powered devices by adapting component power coefficients and aging parameters.

1. **Tablets**: Scale screen power proportionally to display area; reduce radio power contribution due to typically WiFi-dominant usage patterns.

2. **Wearables**: Emphasize idle power and sensor (heart rate, accelerometer) contributions; battery capacities are typically 200–500 mAh with correspondingly scaled parameters.

3. **Electric vehicles**: The aging submodel provides a foundation for battery health estimation systems. The capacity fade model (Equation 25) and cycle-dependent degradation can be scaled to automotive battery packs with appropriate cell-level aggregation.

4. **IoT devices**: For always-on sensors with intermittent connectivity, the radio power model becomes critical; duty cycling optimization directly applies our power decomposition framework.

9 Strengths and Limitations

9.1 Strengths

S-1 **Physically meaningful Shepherd model with rigorous theoretical basis**: The adopted Shepherd model combines theoretical rigor with clear physical interpretation. Each term in the model corresponds to a distinct electrochemical phenomenon: the polarization term ($K/\text{SOC}\cdot I$) captures concentration-dependent overpotential, the ohmic term ($R\cdot I$) represents resistive losses, and the exponential term models initial transient behavior. This physically grounded formulation ensures interpretable parameters and reliable predictions under varying operating conditions.

S-2 **Comprehensive component-level power decomposition**: The explicit separation of power consumption into screen, SoC, radio (4G/5G, WiFi, Bluetooth), GPS, and baseline components enables precise identification of dominant power drivers. This decomposition directly supports targeted optimization strategies for both users and OS designers.

S-3 **Integrated aging dynamics capturing history dependence**: The aging submodel incorporates both cycle aging (power-law with equivalent full cycles N^{γ}) and calendar aging (square-root-of-time $\sqrt{t_s}$), along with Arrhenius temperature dependence and SOC acceleration factors. This unified formulation addresses the problem statement's emphasis on how "a battery's behavior is influenced by its history."

S-4 **Dual-model comparative framework**: The parallel development of both Nernst-Based Model (NBM) and Shepherd model provides different perspectives on battery behavior—NBM offering fundamental electrochemical interpretation while the Shepherd model excels in practical discharge prediction. The systematic comparison between these two distinct approaches validates our model selection and enhances understanding of their respective applicability domains.

S-5 **Validated against real-world datasets**: Rigorous validation using XJTU and Kaggle battery datasets demonstrates superior predictive accuracy, with voltage RMSE below 50 mV and discharge time prediction errors within 3% of experimental observations.

S-6 **Actionable practical recommendations**: The model directly translates into quantitative guidance for users (optimal SOC range, brightness management, GPS restriction) and OS developers (adaptive DVFS, signal-aware radio scaling, age-aware power budgeting).

9.2 Limitations

L-1 **Temperature treated as quasi-static**: While temperature effects on capacity ($Q_0(T)$) and resistance ($R(T)$) are modeled via Arrhenius relationships, the thermal dynamics (heat generation, dissipation, spatial gradients) are not explicitly coupled with the electrical model. This limits accuracy during rapid thermal transients such as gaming bursts.

L-2 **Homogeneous aging assumption**: The model assumes uniform degradation across all cells and electrode regions. In reality, manufacturing variations and localized thermal hotspots cause heterogeneous aging that may induce early cell-to-cell imbalance in multi-cell configurations.

L-3 **Simplified DVFS representation**: The DVFS extension models steady-state power–frequency relationships, omitting transient switching losses and frequency governor dynamics that affect real-time power consumption during rapid workload changes.

L-4 **Constant-power scenario approximation**: Within each usage scenario, power consumption is treated as constant. Realistic usage involves stochastic load fluctuations (variable video bitrates, intermittent GPS fixes) that introduce prediction uncertainty not captured in the deterministic model.

L-5 **Self-discharge neglected**: For extended standby periods exceeding several days, self-discharge mechanisms (estimated at 2–5% per month for Li-ion cells) may become non-negligible and should be incorporated for long-term predictions.

10 Conclusion

We model smartphone battery drain using several approaches. Rather than having a single, prescriptive metric, we come up with several which correspond to battery behavior in different ways. We then select a subset of these predictors through systematic model comparison.

By using the Nernst-Based Model (NBM) as a simple yet effective representation of electrochemical thermodynamics, we are able to condense the information about the voltage–SOC relationship into interpretable parameters, achieving reasonable accuracy with minimal complexity. Building upon this foundation, we develop the Shepherd model which provides comprehensive modeling of battery dynamics and achieves highly accurate predictions, capturing voltage plateau characteristics and current-dependent polarization with superior precision.

Using an innovative physics-based framework, we are able to predict how long a battery will last under various usage conditions, how battery aging affects performance over time, and how temperature influences discharge behavior. The innovative aspect of the model is that it is made up of three submodels (voltage dynamics, power decomposition, and aging effects) which, when conditioned upon the data, are independent. Hence the overall model can be quite large while still being efficient enough to perform numerical integration and obtain predictions from the posterior distribution. Given the physics-based nature of the model, it is also able to provide uncertainties on any predictions made in the form of parameter sensitivities instead of simple point estimates. We provide several predictions supposing representative usage scenarios and provide several validations to confirm that our model aligns with experimental data.

References

[1] H. Ali, H. A. Khan, and M. G. Pecht, *Evaluation of li-based battery current, voltage, and temperature profiles for in-service mobile phones*, IEEE Access, 8 (2020), pp. 73665–73676.

[2] E. Benkhelifa, T. Welsh, L. Tawalbeh, Y. Jararweh, and A. Basalamah, *Energy optimisation for mobile device power consumption: A survey and a unified view of modelling for a comprehensive network simulation*, Mobile Networks and Applications, 21 (2016), pp. 575–588.

[3] S. Benson, A. Lele, and K. Arora, *Predicting the life of li-ion batteries using the arrhenius model*, ECS Transactions, 41 (2012), pp. 87–94.

[4] G. Bhat, S. Gumussoy, and U. Y. Ogras, *Power-temperature stability and safety analysis for multiprocessor systems*, ACM Trans. Embed. Comput. Syst., 16 (2017), p. 145.

[5] G. Bhat, S. K. Mandal, S. T. Manchukonda, S. V. Vadlamudi, A. Agarwal, J. Wang, and U. Y. Ogras, *Per-core power modeling for heterogenous socs*, Electronics, 10 (2021), p. 2428.

[6] C. R. Birkl, E. McTurk, M. R. Roberts, P. G. Bruce, and D. A. Howey, *A parametric open circuit voltage model for lithium ion batteries*, J. Electrochem. Soc., 162 (2015), pp. A2271–A2280.

[7] A. Carroll and G. Heiser, *An analysis of power consumption in a smartphone*, in Proc. USENIX Annual Technical Conference, 2010, p. 21.

[8] X. Chen, Y. Chen, M. Dong, and C. Zhang, *Demystifying energy usage in smartphones*, in Proc. 51st Annual Design Automation Conference (DAC), 2014, pp. 1–5.

[9] Y. Chen and K. Mitsubori, *Dependence of ocv, hysteresis, overvoltage on temperature and soc in commercial lifepo4 lithium-ion battery*, Electronics Letters, 61 (2025).

[10] A. El Outmani, E. M. Jaara, and M. Azizi, *Literature review of energy consumption modeling for mobile devices*, in Proc. Int. Conf. Advanced Intelligent Systems for Sustainable Development, 2023, pp. 99–112.

[11] Z. He, D. Guo, X. Liu, and G. Yang, *An evaluation method of battery dc resistance consistency caused by temperature variation*, in Proc. IEEE IECON, 2017, pp. 7623–7628.

[12] X. Hu, S. Li, and H. Peng, *A comparative study of equivalent circuit models for li-ion batteries*, J. Power Sources, 198 (2012), pp. 359–367.

[13] e. a. Kou, *Experimental study on factors affecting internal resistance of lithium power batteries*, Automotive Engineering, 39 (2017), p. 7.

[14] M. Lauridsen, L. Noel, T. B. Sørensen, and P. E. Mogensen, *An empirical lte smartphone power model with a view to energy efficiency evolution*, Intel Technology Journal, 18 (2014), pp. 172–193.

[15] X. Li, Q. Wang, Y. Yang, and J. Kang, *Correlation between capacity loss and measurable parameters of lithium-ion batteries*, Int. J. Electrical Power & Energy Systems, 110 (2019), pp. 819–826.

[16] M.-H. M. Lu, M. Hack, R. Hewitt, M. S. Weaver, and J. J. Brown, *Power consumption and temperature increase in large area active-matrix oled displays*, J. Display Technology, 4 (2008), pp. 47–53.

[17] R. Murmuria, J. Medsger, A. Stavrou, and J. M. Voas, *Mobile application and device power usage measurements*, in Proc. IEEE Int. Conf. Software Security and Reliability, 2012, pp. 147–156.

[18] L. Negri and D. Zanetti, *Power/performance tradeoffs in bluetooth sensor networks*, in Proc. 39th Hawaii Int. Conf. System Sciences (HICSS), vol. 9, 2006, pp. 236b–236b.

[19] Y. Sun, J. Chen, Y. Tang, and Y. Chen, *Energy modeling of iot mobile terminals on wifi environmental impacts*, Sensors, 18 (2018), p. 1728.

[20] J. Vetter, P. Novák, M. Wagner, C. Veit, K.-C. Möller, J. Besenhard, M. Winter, M. Wohlfahrt-Mehrens, C. Vogler, and A. Hammouche, *Ageing mechanisms in lithium-ion batteries*, J. Power Sources, 147 (2005), pp. 269–281.

[21] D. Xu, A. Zhou, X. Zhang, G. Wang, X. Liu, C. An, Y. Shi, L. Liu, and H. Ma, *Understanding operational 5g: A first measurement study on its coverage, performance and energy consumption*, in Proc. ACM SIGCOMM, 2020, pp. 479–494.

[22] H. W. You, J. I. Bae, S. J. Cho, J. M. Lee, and S.-H. Kim, *Analysis of equivalent circuit models in lithium-ion batteries*, AIP Advances, 8 (2018).

[23] J. Yu, Y. Guo, and W. Zhang, *Quantitative analysis method for full lifecycle aging pathways of lithium-ion battery systems based on equilibrium potential reconstruction*, Applied Sciences, 15 (2025), p. 10079.

Report on Use of AI

We utilized Large Language Models (LLMs), specifically ChatGPT (GPT-4) and GitHub Copilot, as auxiliary tools during the preparation of this paper. The AI tools were employed in the following capacities, with all outputs carefully reviewed, validated, and modified by team members to ensure accuracy and appropriateness.

Translation Assistance

We used LLMs for translation purposes to convert technical descriptions and literature summaries between Chinese and English, ensuring accurate terminology usage in battery electrochemistry and power electronics domains.

Query: "Translate the following technical paragraph about lithium-ion battery SEI layer formation from Chinese to English, maintaining precise electrochemical terminology"

> AI provided accurate translations of technical content related to solid electrolyte interphase (SEI) layer growth, Arrhenius temperature dependence, and state of charge dynamics. The translations preserved domain-specific terminology such as "intercalation," "coulombic efficiency," and "concentration polarization." Team members with bilingual proficiency verified all translations for technical accuracy.

Code Development and Improvement

Query: "Optimize Python code for numerical integration of battery SOC dynamics with adaptive step size"

> AI assisted in refactoring simulation code for improved computational efficiency, suggested using `scipy.integrate.solve_ivp` with event detection for cutoff conditions, and helped debug numerical instabilities near SOC boundaries. All generated code was manually reviewed and validated against analytical solutions before use.

Query: "Write a Python function to implement the Nernst-based OCV-SOC model with exponential correction term"

> AI generated initial code structure implementing Equation 9, including proper handling of logarithmic singularities as SOC approaches 0 or 1. Team members modified the code to include temperature dependence and validated outputs against manufacturer discharge curves.

Query: "Debug this error: RuntimeWarning: divide by zero encountered in log"

> AI identified that the Nernst term $\ln(\mathrm{SOC}/(1-\mathrm{SOC}))$ produces numerical overflow near boundary conditions and suggested implementing SOC clamping with $\epsilon = 10^{-6}$ tolerance. This solution was adopted after verifying that it introduced negligible error ($< 0.01\%$) in voltage predictions.

Query: "Generate matplotlib code to create 3D surface plot of voltage as function of SOC and current"

AI provided visualization code using `matplotlib.pyplot` and `mpl_toolkits.mplot3d` for generating Figure 3. Team members customized color maps, axis labels, and viewing angles to improve clarity of the voltage landscape presentation.

Model Development Assistance

Query: "Refine continuous-time battery model equations with proper SoC notation and physical justification for aging effects"

AI suggested standardizing "soc" to "SoC" for consistency with battery literature, recommended including exponential voltage term for open-circuit voltage nonlinearity, and proposed modeling aging through cumulative charge throughput $N(t)$ with first-order degradation kinetics for capacity and resistance—consistent with SEI growth mechanisms in lithium-ion batteries.

Query: "Explain the physical basis for the Shepherd battery model and its limitations"

AI provided background on the Shepherd model's empirical origins and its decomposition into polarization, ohmic, and transient voltage components. This helped frame Section 3.2 and clarify why the Nernst-based approach offers improved physical interpretability while maintaining computational simplicity.

Query: "How should capacity fade and resistance growth be coupled in a battery aging model?"

AI suggested that the linear correlation between capacity loss and resistance increase (as reported in [15]) could be exploited to reduce model complexity by expressing Q_0 as a function of R. This insight led to the formulation of Equation 25 and ensured internal consistency of the aging submodel.

Literature Review Support

Query: "Summarize key findings from recent papers on smartphone battery power consumption modeling"

AI provided structured summaries of referenced papers, helping identify relevant methodologies and parameter ranges. All cited sources were independently verified by team members through direct reading of original publications.

Query: "What are the typical values of activation energy for lithium-ion battery aging?"

AI indicated that activation energies for SEI growth typically range from 40–60 kJ/mol based on literature values. This information was cross-referenced with [20] and [3] to select $E_a = 50,000$ J/mol for our model parameters in Table 1.

Query: "Compare equivalent circuit models versus electrochemical models for battery state estimation"

> AI summarized trade-offs between model fidelity and computational cost, noting that equivalent circuit models (Thevenin, RC networks) offer fast computation but limited physical insight, while electrochemical models (pseudo-2D, single particle) provide mechanistic understanding at higher computational expense. This comparison informed our choice of the Nernst-based approach as a middle ground.

Writing and Formatting

Query: "Improve clarity of technical descriptions for battery aging mechanisms"

> AI assisted in refining sentence structure, improving technical clarity, and ensuring consistent notation throughout the manuscript. LaTeX formatting suggestions were also provided for tables and equations.

Query: "Suggest a clear structure for presenting model validation results"

> AI recommended organizing validation into progressive sections: dataset overview, accuracy comparison, boundary behavior analysis, and ablation studies. This structure was adopted for Section 5 with modifications to emphasize the structural advantages of our physics-based approach.

Query: "How to format a LaTeX table with merged cells for battery parameter comparison?"

> AI provided LaTeX code examples using `multirow` and `multicolumn` packages for creating the parameter summary table. Formatting was adjusted by team members to match journal style guidelines.

Query: "Proofread this paragraph for grammar and academic tone"

> AI identified grammatical errors, suggested more precise word choices (e.g., "degradation" instead of "decay" in certain contexts), and flagged potential ambiguities in technical descriptions. All suggestions were reviewed and selectively adopted based on team judgment.

Data Analysis Support

Query: "How to perform sensitivity analysis for a multi-parameter battery model?"

> AI suggested using one-at-a-time (OAT) sensitivity analysis for initial screening followed by Sobol indices for global sensitivity assessment. The OAT approach was implemented for the results presented in Section 5, with parameter ranges based on literature values.

Query: "Calculate the relative contribution of each voltage term at different SOC levels"

AI helped structure the voltage decomposition analysis presented in Equation 11, providing numerical examples that were verified by independent calculation. This analysis revealed that electrochemical effects dominate at low SOC while ohmic losses dominate at mid-range SOC.

Limitations of AI Assistance

During our use of AI tools, we encountered several limitations that required human intervention:

1. AI occasionally provided outdated parameter values from older publications; all numerical values were verified against recent (2020+) literature.
2. AI-generated code sometimes contained subtle logical errors that only became apparent during validation; systematic testing was essential.
3. AI suggestions for model structure occasionally prioritized mathematical elegance over physical plausibility; domain expertise was required to evaluate trade-offs.
4. Translation of highly specialized terms (e.g., specific battery chemistries) sometimes required manual correction based on current industry terminology.

Statement of Responsibility

All mathematical derivations, physical interpretations, modeling decisions, and conclusions presented in this paper are the original work of the team members. AI tools served as assistants for language refinement, code debugging, literature summarization, and organizational tasks. The team independently verified all AI-generated content, made final decisions on model formulation, and takes full responsibility for the accuracy and validity of all content presented in this paper.

The use of AI in this work adhered to ethical guidelines for academic integrity. AI was not used to generate novel scientific claims or to fabricate data. All substantive intellectual contributions—including the Nernst-based OCV-SOC formulation, the coupled aging model, and the practical recommendations—originated from team members' analysis and judgment.